

# Angular momentum bookkeeping IDENTITY residual (not a conservation test)

Escaped trajectories

1  
0.8  
0.6  
0.4  
0.2  
0

-14

-12

-10

-8

-6

-4

-2

0

$\log_{10}(|\Delta(J_N + J_{\text{rad}})|/J_{\text{scale}})$

Entries: N = 1

Fit: degenerate Gaussian ( $\sigma=0$ )

$\mu = -7.867$

$\sigma = 0$

IDENTITY by construction:  $E_{\text{bound}}$  is defined

as the residual that closes this sum;

this measures roundoff + endpoint interpolation.

Independent physical residual:

$|E_{\text{bound}}|/E_{\text{rad}}$  median/p95 = 30.304 / 30.304